

Unit 23: Antiderivatives

1. The idea, in plain words

Every skill so far has gone one direction: start with a function, find its derivative. This unit runs the movie backward. Given a derivative, can you figure out what function it came from? That “undoing” process is called finding an antiderivative.

Picture it like this: if differentiation is “what’s my speed, given my position?”, antiderivatives ask the reverse question: “given my speed the whole time, what was my position?” You already have great intuition for this from Unit 11 — position, velocity, and acceleration are all connected by derivatives, so they must also be connected by antiderivatives, just read in the opposite direction.

Here’s the catch that makes antiderivatives a little different from derivatives: they’re never unique. The derivative of x^2 is $2x$ — but so is the derivative of $x^2 + 5$, or $x^2 - 100$, or x^2 plus any constant, since a constant’s derivative is always 0. So when you’re asked to find “the” antiderivative of $2x$, the honest answer is a whole family of functions: $x^2 + C$, where C can be any constant. We always tack on that mystery constant, written $+C$, to represent every possible antiderivative at once.

If you’re given extra information — like the function’s value at one specific point — you can pin down exactly which member of that family you want, by solving for the specific value of C that makes it fit.

2. Toolbox

Notation: the antiderivative of $f(x)$ is written $\int f(x) dx$, and read “the indefinite integral of f .” If $F'(x) = f(x)$, then:

$$\int f(x) dx = F(x) + C$$

Power Rule for antiderivatives (the reverse of the power rule for derivatives): bump the exponent up by one, then divide by the new exponent.

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (\text{for } n \neq -1)$$

(The case $n = -1$, i.e., $\int \frac{1}{x} dx$, needs a special function — natural logarithms — which we haven’t covered yet. We’ll come back to it in a later unit.)

Constant Multiple and Sum Rules (same as for derivatives):

$$\int c \cdot f(x) dx = c \int f(x) dx \quad \int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$$

Basic trig antiderivatives:

$$\begin{aligned} \int \sin x dx &= -\cos x + C & \int \cos x dx &= \sin x + C \\ \int \sec^2 x dx &= \tan x + C & \int \csc^2 x dx &= -\cot x + C \end{aligned}$$

$$\int \sec x \tan x \, dx = \sec x + C \qquad \int \csc x \cot x \, dx = -\csc x + C$$

Solving for C using extra information (an “initial condition”): find the general antiderivative first (with $+C$ still attached), then plug in the given point to solve for the exact value of C .

Chaining antiderivatives together (for motion problems): since $a(t) = v'(t)$ and $v(t) = s'(t)$, you can work backward: antidifferentiate acceleration to get velocity (using $v(0)$ to solve for the constant), then antidifferentiate that velocity to get position (using $s(0)$ to solve for the next constant).

3. Common mistakes

- Forgetting the $+C$. This is the single most common error in this entire unit — every single indefinite integral needs it, even if the problem seems simple.
- Sign errors with $\sin x$ and $\cos x$. Remember the antiderivative of $\sin x$ is $-\cos x$ (with a negative sign) — it’s easy to drop that minus sign, especially since the derivative rules go the opposite direction.
- Applying the power rule to x^{-1} . This case is special (it needs natural logs, covered later) — don’t try to force the regular power rule formula onto it, since dividing by $n + 1 = 0$ is undefined anyway.
- Forgetting to distribute the antiderivative across every term in a sum, or forgetting a term entirely when there are several to keep track of.
- Forgetting to actually solve for C in an initial-value problem. Finding the general antiderivative with a leftover $+C$ isn’t the final answer if you were given a specific point — you must use that point to pin down the exact value of C .

4. Problem Set

Warm-up

1. $\int x^3 \, dx$
2. $\int 5 \, dx$
3. $\int x^4 \, dx$
4. $\int 3x^2 \, dx$
5. $\int (x^2 + 2x) \, dx$
6. $\int (4x^3 - 3x^2 + 1) \, dx$
7. $\int \sqrt{x} \, dx$

Standard

8. $\int \sin x \, dx$
9. $\int (3\cos x - 2\sin x) \, dx$

10. $\int \sec^2 x \, dx$
11. $\int \frac{2}{x^3} \, dx$
12. $\int \left(x^2 - \frac{1}{x^2} \right) \, dx$
13. $\int (\sec x \tan x + \csc x \cot x) \, dx$
14. $\int (4x^3 - 6\sqrt{x}) \, dx$

Challenge

15. Find $f(x)$ given $f'(x) = 3x^2 - 4x + 1$ and $f(0) = 5$.
16. Find $f(x)$ given $f'(x) = 6x^2 - 2$ and $f(1) = 3$.
17. A particle's acceleration is $a(t) = 6t - 4$. Given $v(0) = 3$ and $s(0) = 2$, find $v(t)$ and $s(t)$.
18. Find $f(x)$ given $f'(x) = 2\sin x + 3\cos x$ and $f(0) = 4$.
19. $\int (x+1)(x-2) \, dx$

Applied

20. A ball is thrown upward with acceleration due to gravity $a(t) = -32 \text{ ft/s}^2$, initial velocity $v(0) = 48 \text{ ft/s}$, and initial height $s(0) = 0$. Find $v(t)$ and $s(t)$, then find the ball's maximum height.
21. A company's marginal cost is $MC(x) = 3x^2 - 12x + 15$ dollars per unit, and fixed costs are \$200 (i.e., $C(0) = 200$). Find the total cost function $C(x)$.
22. A company's marginal revenue is $MR(x) = 50 - 2x$, with $R(0) = 0$. Find the revenue function $R(x)$.
23. A car brakes with acceleration $a(t) = -10 \text{ ft/s}^2$, starting from velocity $v(0) = 60 \text{ ft/s}$. Find $v(t)$, and determine how long it takes the car to come to a complete stop.